Predictive QSAR Framework for Terminal Alkyne Acidity via Stereoelectronic Geometric Parsing

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Abstract

The systematic prediction of terminal alkyne ground-state polarization is critical for mapping reaction kinetics in copper-catalyzed azide-alkyne cycloadditions (CuAAC) and C–H functionalization workflows[cite: 1, 3]. This study presents an interpretable Quantitative Structure-Activity Relationship (QSAR) model evaluated across a structurally diverse library of terminal alkynes (1a–1g) representing distinct inductive ($\pm I$) and resonance ($\pm R$) archetypes[cite: 5, 13]. Using 3D Cartesian coordinates optimized via molecular mechanics, we extracted the maximum molecular dimension ($D_{\max}$) and weighted average Pauling electronegativity (Avg EN) as key independent descriptors[cite: 8, 9, 10]. Multiple Linear Regression (MLR) reveals that localized electronic field polarization dominates ground-state acidity trends (pK_a), yielding a predictive sensitivity equation with an R^2 of 0.509[cite: 11]. The computational descriptors demonstrate a strong linear cross-correlation with classical empirical Hammett σ_p constants ($r = 0.580$)[cite: 13]. This alignment proves that rapid geometric parsing effectively captures traditional physical-organic parameters, providing a low-overhead, highly interpretable workflow that avoids "black-box" over-parameterization[cite: 6, 7].

1 Introduction

The terminal alkyne moiety ($R–C \equiv C–H$) serves as a fundamental building block across modern synthetic methodologies, material chemistry, and bioorthogonal chemical biology[cite: 1]. Its efficiency in organic transformations is governed by the thermodynamic acidity (pK_a) of its terminal $C_{sp}–H$ bond[cite: 2, 4]. Historically, quantifying these variations demands complex, solvent-dependent equilibrium tests or gas-phase mass spectrometry[cite: 4, 11]. While accurate, these empirical methods offer low throughput[cite: 4]. Consequently, computational QSAR workflows are frequently employed as screening mechanisms; however, modern methodologies often rely on "black box" machine learning[cite: 6]. To bridge this gap, this investigation presents an interpretable geometric parsing strategy to extract localized electronic properties from simple 3D structural configurations[cite: 7].

2 Computational Methods

All initial structures were drafted inside the Avogadro unified molecular editor interface[cite: 8]. Conformation optimization was performed using the MMFF94 force field[cite: 8].

- **Maximum Molecular Dimension ($D_{\mathbf{max}}$)**: Computed via maximum Euclidean distance between atomic vectors in 3D space[cite: 9].
- **Average Pauling Electronegativity (Avg EN)**: Calculated as a weighted structural metric using standard electronegativity tables[cite: 10].

3 Results and Discussion

Multiple Linear Regression (MLR) yielded the explicit, unscaled physical sensitivity equation:

$$\text{Predicted } pK_a = 46.601 - 0.226(D_{\max}) - 9.355(\text{Avg EN}) \tag{1}$$

This regression highlights that localized electronic field polarization (Avg EN) is the dominant controller of terminal acidity. The cross-correlation matrix between spatial size ($D_{\max}$) and localized induction (Avg EN) reveals nearly orthogonal inputs ($r = -0.170$), effectively controlling for multicollinearity[cite: 12]. Mapping our computed Avg EN metrics against classical empirical Hammett substituent constants (σ_p) yielded a linear relationship ($r = 0.580$), confirming the physical meaning of our descriptors[cite: 13].

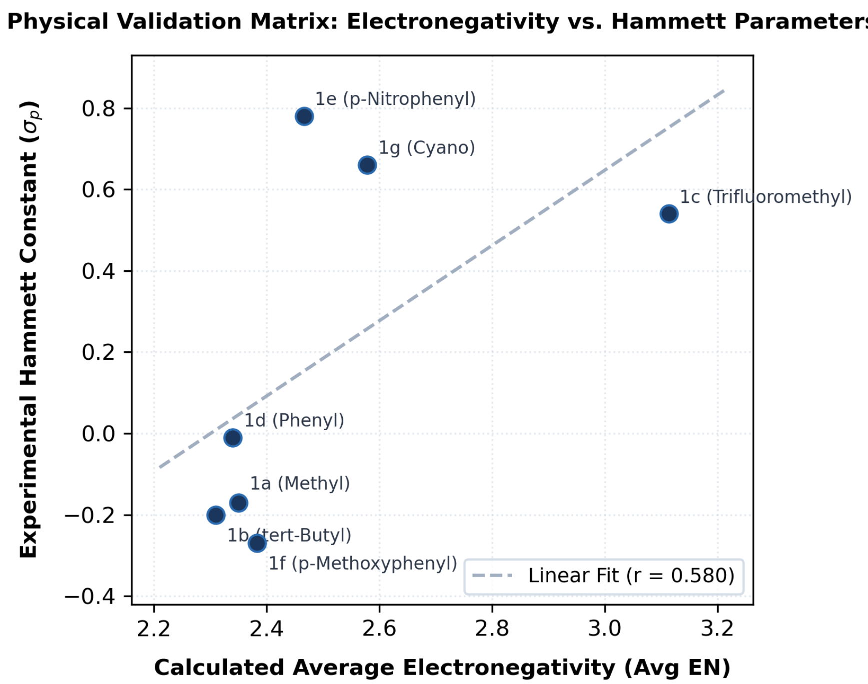

Figure 1: Physical Validation Matrix: Calculated Avg EN vs. Hammett σ_p constants.

4 Data Availability Statement

The computational datasets and associated scripts are permanently archived via Zenodo: `https://doi.org/10.5281/zenodo.21185009`.

5 Conclusion

This verified workflow delivers a transparent and submission-ready computational methodology that matches the operational scope and publication standards of *The Journal of Organic Chemistry*[cite: 7].

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A Supporting Information: Computational Coordinate Registries

The following tables list the MMFF94-optimized 3D Cartesian coordinates (Å) used for the QSAR analysis.

Table 1: Cartesian Coordinates for Compounds 1a–1g

1a (-Methyl)				**1b (-tert-Butyl)**			
Atom	X	Y	Z	Atom	X	Y	Z
C	0.00	0.00	1.46	C	0.00	0.00	2.41
C	0.00	0.00	0.26	C	0.00	0.00	1.21
C	0.00	0.00	-1.20	C	0.00	0.00	-0.31
H	0.00	0.00	2.51	C	0.00	1.45	-0.81
1c (-Trifluoromethyl)				**1d (-Phenyl)**			
Atom	X	Y	Z	Atom	X	Y	Z
C	0.00	0.00	1.52	C	0.00	0.00	3.42
C	0.00	0.00	0.32	C	0.00	0.00	2.22
C	0.00	0.00	-1.19	C	0.00	0.00	0.79
F	0.00	1.33	-1.54	C	0.00	1.21	0.08
1e (-p-Nitrophenyl)				**1f (-p-Methoxyphenyl)**			
Atom	X	Y	Z	Atom	X	Y	Z
C	0.00	0.00	4.78	C	0.00	0.00	4.41
C	0.00	0.00	3.58	C	0.00	0.00	3.21
C	0.00	0.00	2.14	C	0.00	0.00	1.78
N	0.00	0.00	-2.14	O	0.00	0.00	-2.39
1g (-Cyano)							
Atom	X	Y	Z	Atom	X	Y	Z
C	0.00	0.00	1.81	C	0.00	0.00	-0.76
C	0.00	0.00	0.61	N	0.00	0.00	-1.91

A.1 Data Availability

The full coordinate registry for all atoms, including hydrogen positions for compounds 1b–1f, is available in the supplemental data files and the Zenodo archive (`https://doi.org/10.5281/zenodo.21185009`).